1)

A: $2 + 2\mathrm{i}$

B: $6 + 5\mathrm{i}$

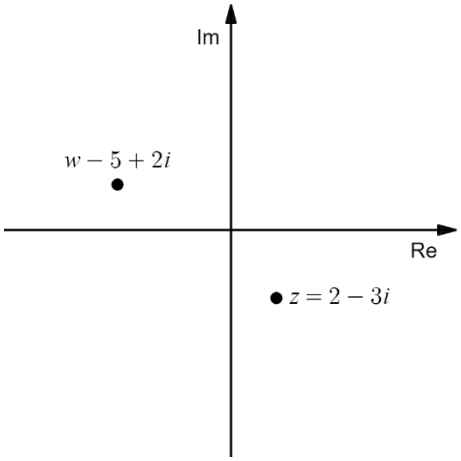
C: $-4 + 2\mathrm{i}$

D: $-4 - 2\mathrm{i}$

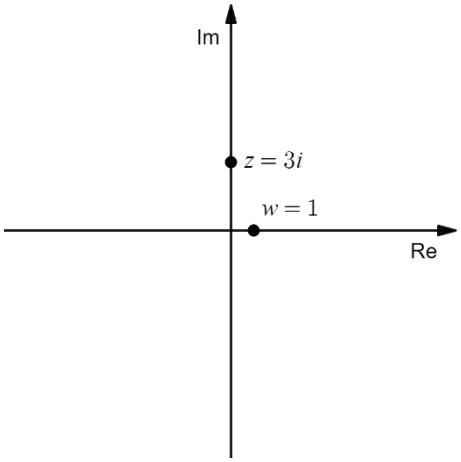
E: $-9 + 6\mathrm{i}$

F: $6 - 4\mathrm{i}$

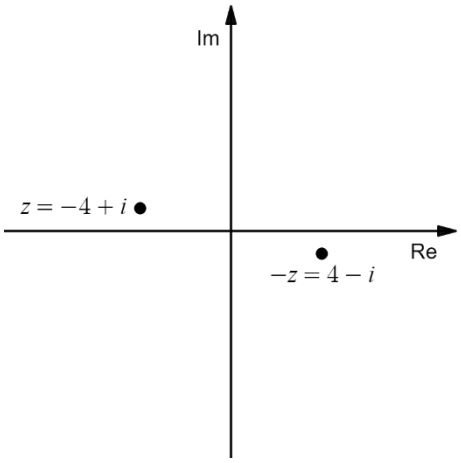
2) a)



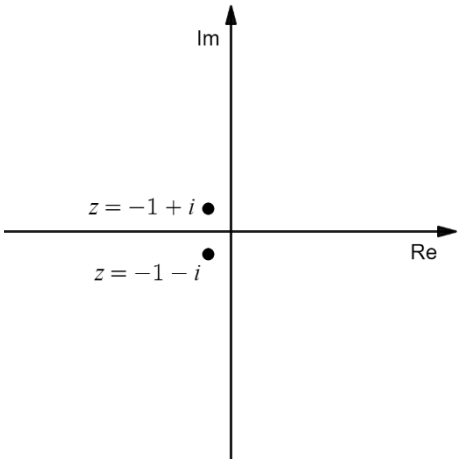
2) b)



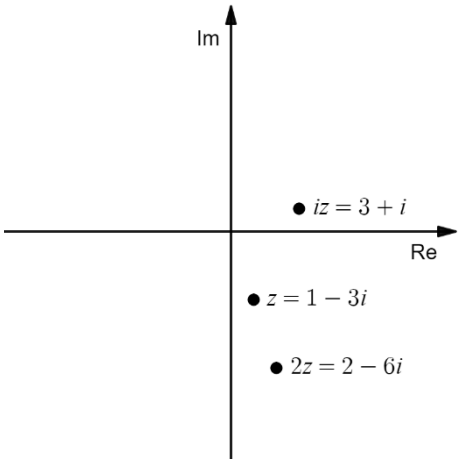
2) c)



2) d)

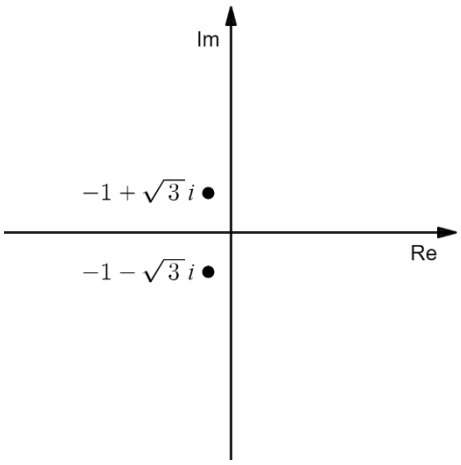


2) e)



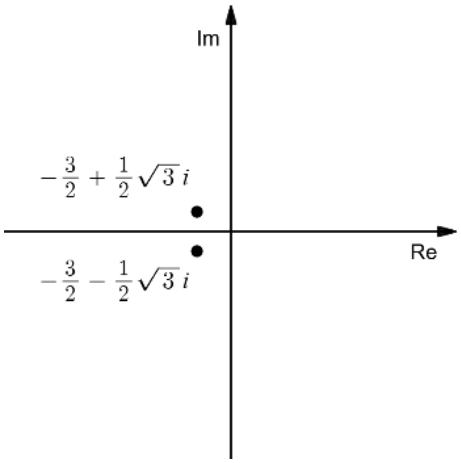
3) a)

$z^2 + 2z + 4 = 0$ has solutions $z = -1 \pm \sqrt{3}\text{i}$



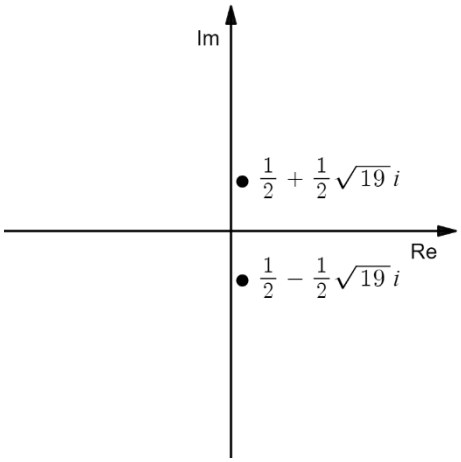
3) b)

$z^2 - 3z + 3 = 0$ has solutions $z = \frac{3}{2} \pm \frac{1}{2}\sqrt{3}\text{i}$



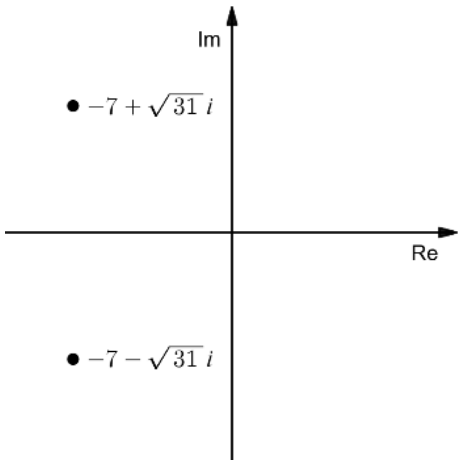
3) c)

$z^2 - z + 5 = 0$ has solutions $z = \frac{1}{2} \pm \frac{1}{2}\sqrt{19}\text{i}$



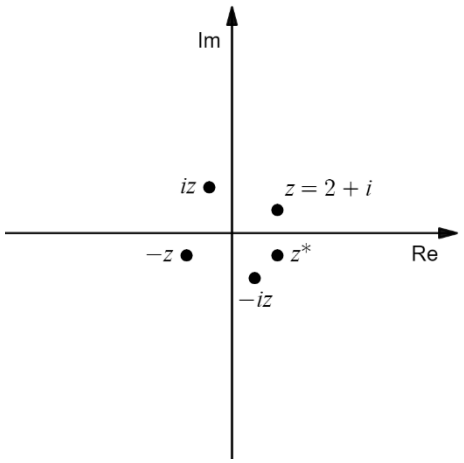
3) d)

$z^2 + 14z + 80 = 0$ has solutions $z = -7 \pm \sqrt{31}\text{i}$



4)

$z = 2 + \text{i}$, so that $z^* = 2 - \text{i}$, $-z = -2 - \text{i}$, $\text{i}z = -1 + 2\text{i}$, $-\text{i}z = 1 - 2\text{i}$



4) a)

$z^* = 2 - \text{i}$

Reflection across the real axis

4) b)

$-z = -2 - \text{i}$

Rotation by 180° about the origin

4) c)

$\text{i}z = -1 + 2\text{i}$

Rotation by 90° anticlockwise about the origin (or 270° clockwise)

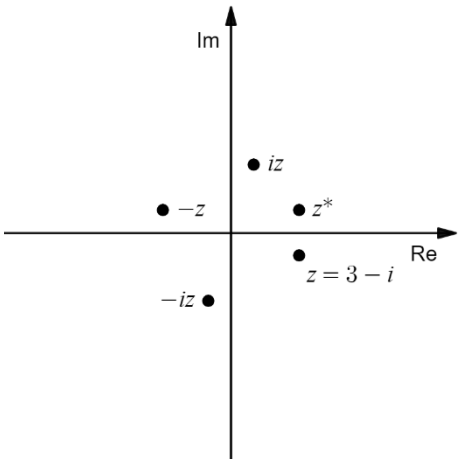
4) d)

$-\mathrm{i}z = 1 - 2\mathrm{i}$

Rotation by $270°$ anticlockwise about the origin (or $90°$ clockwise)

5)

$z = 3 - \mathrm{i}$, so that $z^* = 3 + \mathrm{i}$, $-z = -3 + \mathrm{i}$, $\mathrm{i}z = 1 + 3\mathrm{i}$, $-\mathrm{i}z = -1 - 3\mathrm{i}$



5) a)

$z^* = 3 + \mathrm{i}$

Reflection across the real axis

5) b)

$-z = -3 + \mathrm{i}$

Rotation by $180°$ about the origin

5) c)

$\mathrm{i}z = 1 + 3\mathrm{i}$

Rotation by $90°$ anticlockwise about the origin (or $270°$ clockwise)

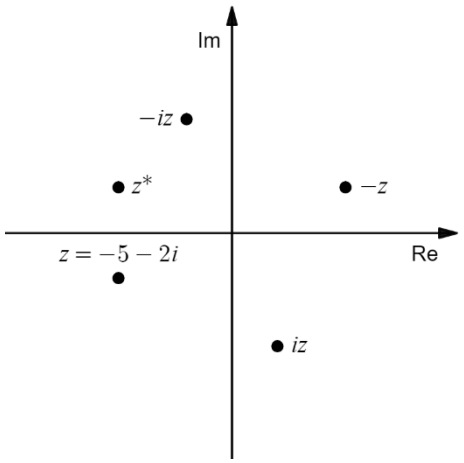
5) d)

$-\mathrm{i}z = -1 - 3\mathrm{i}$

Rotation by $270°$ anticlockwise about the origin (or $90°$ clockwise)

6) a) b) c) d)

$z = -5 - 2\text{i}$, so that $z^* = -5 + 2\text{i}$, $-z = 5 + 2\text{i}$, $\text{i}z = 2 - 5\text{i}$, $-\text{i}z = -2 + 5\text{i}$



6) a)

$z^* = -5 + 2\text{i}$

Reflection across the real axis

6) b)

$-z = 5 + 2\text{i}$

Rotation by $180°$ about the origin

6) c)

$\text{i}z = 2 - 5\text{i}$

Rotation by $90°$ anticlockwise about the origin (or $270°$ clockwise)

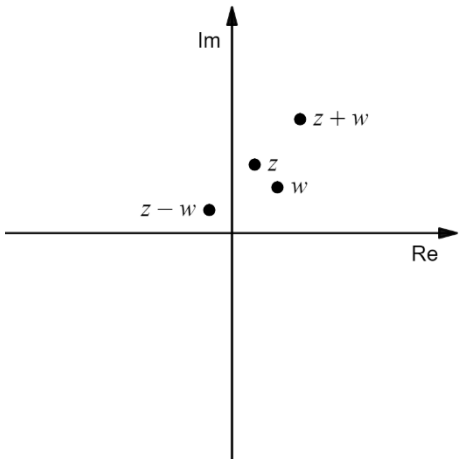
6) d)

$-\text{i}z = -2 + 5\text{i}$

Rotation by $270°$ anticlockwise about the origin (or $90°$ clockwise)

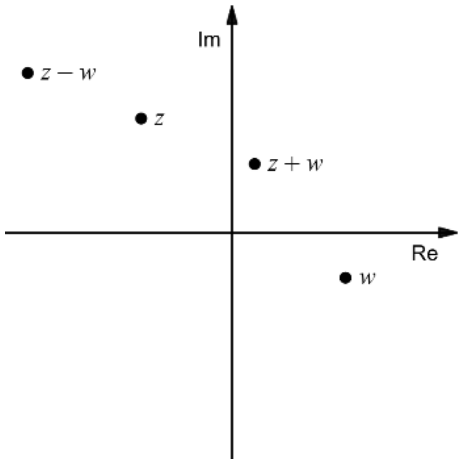
7) a) b) c) d)

$z + w = 3 + 5\text{i}$ and $z - w = -1 + \text{i}$



8) a) b) c) d)

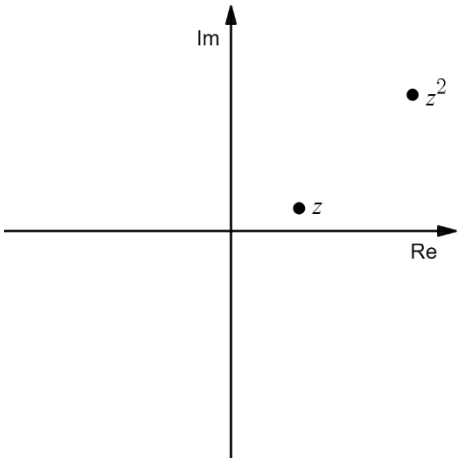
$z + w = 1 + 3\text{i}$ and $z - w = -9 + \text{i}$



9) a)

$z^2 = (3 + \text{i})(3 + \text{i}) = 9 + 6\text{i} - 1 = 8 + 6\text{i}$ as req.

9) b)



10) a)

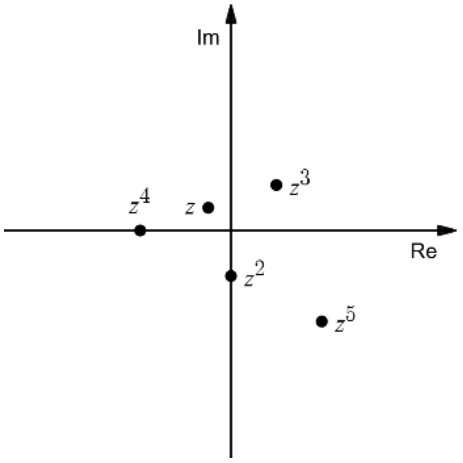
$z^2 = (-1 + \mathrm{i})(-1 + \mathrm{i}) = 1 - 2\mathrm{i} - 1 = -2\mathrm{i}$

$z^3 = (-1 + \mathrm{i})(-2\mathrm{i}) = 2 + 2\mathrm{i}$

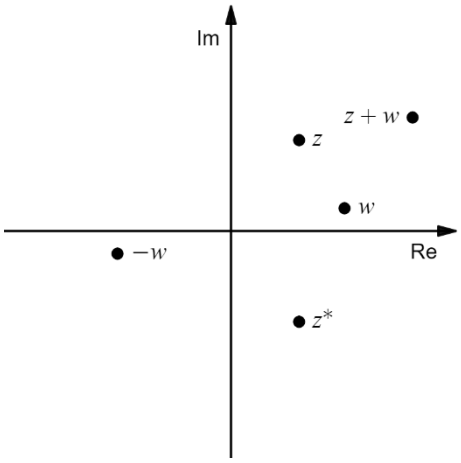
$z^4 = (-1 + \mathrm{i})(2 + 2\mathrm{i}) = -2 - 2\mathrm{i} + 2\mathrm{i} - 2 = -4$

$z^5 = -4(-1 + \mathrm{i}) = 4 - 4\mathrm{i}$

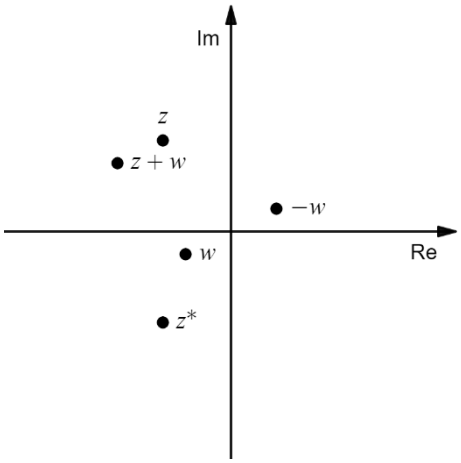
10) b)



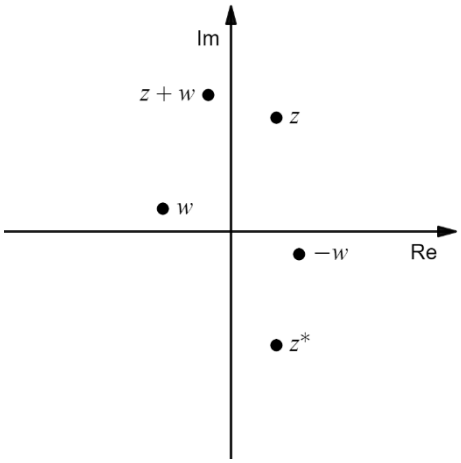
11) a)



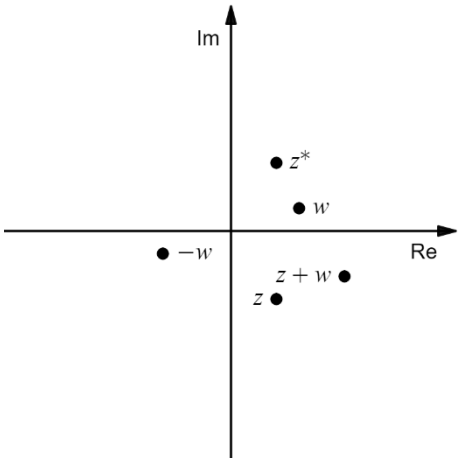
11) b)



11) c)



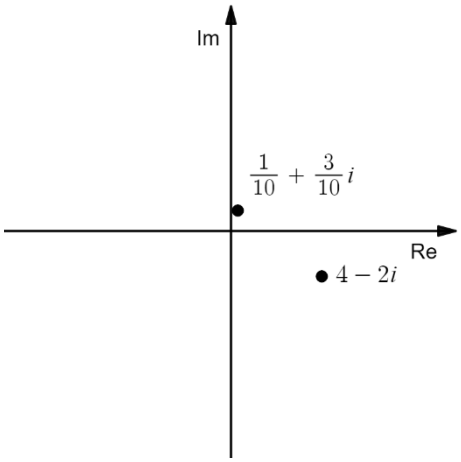
11) d)



12) a)

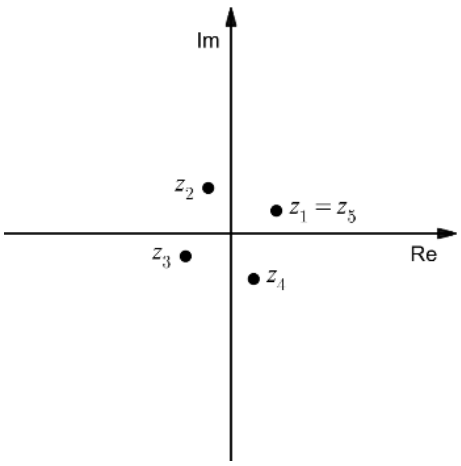
$$z_2 = \frac{1+\mathrm{i}}{4-2\mathrm{i}} \times \frac{4+2\mathrm{i}}{4+2\mathrm{i}} = \frac{4+6\mathrm{i}-2}{16+4} = \frac{2+6\mathrm{i}}{20} = \frac{1}{10} + \frac{3}{10}\mathrm{i}$$

12) b)



13) a)

$z_2 = \mathrm{i}(2 + \mathrm{i}) = -1 + 2\mathrm{i}$
$z_3 = -(2 + \mathrm{i}) = -2 - \mathrm{i}$
$z_4 = -\mathrm{i}(2 + \mathrm{i}) = 1 - 2\mathrm{i}$
$z_5 = 1(2 + \mathrm{i}) = 2 + \mathrm{i}$

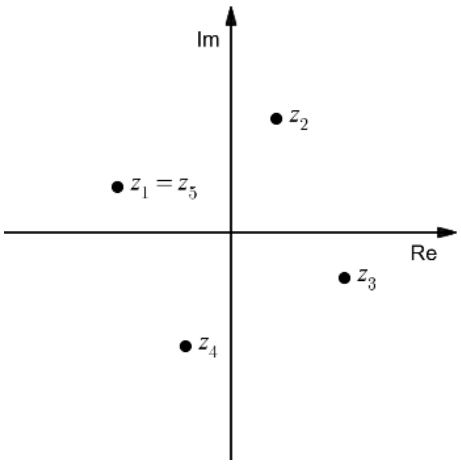


13) b)

Rotation of 90° anti-clockwise (or 270° clockwise) about the origin.

14) a)

$z_2 = -\mathrm{i}(-5 + 2\mathrm{i}) = 2 + 5\mathrm{i}$
$z_3 = -\mathrm{i}(2 + 5\mathrm{i}) = 5 - 2\mathrm{i}$
$z_4 = -\mathrm{i}(5 - 2\mathrm{i}) = -2 - 5\mathrm{i}$
$z_5 = -\mathrm{i}(-2 - 5\mathrm{i}) = -5 + 2\mathrm{i}$

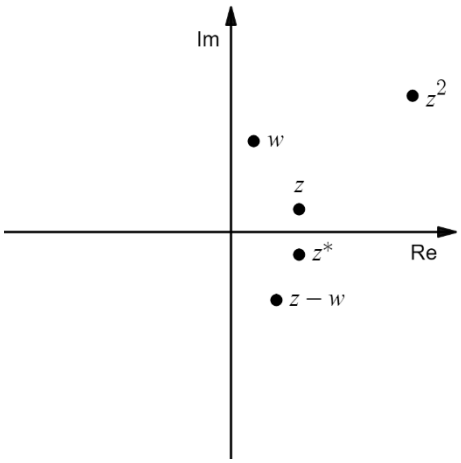


14) b)

Rotation by 90° clockwise (or 270° anti-clockwise) about the origin

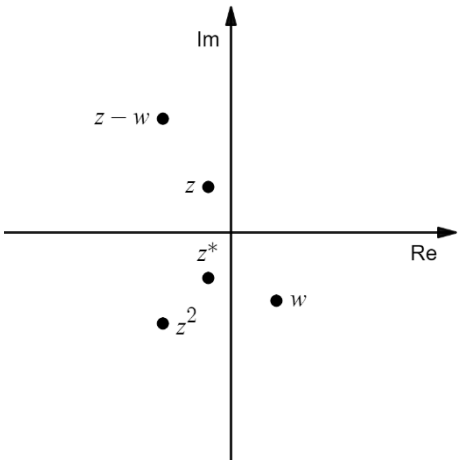
15) a)

$(a + b\text{i})^2 = a^2 - b^2 + 2ab\text{i}$ since $a > b$, z^2 is in top right quadrant.

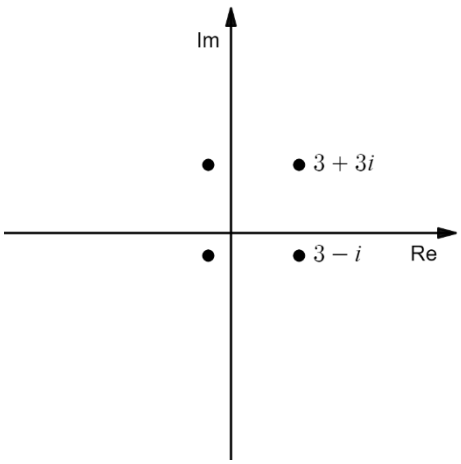


15) b)

$(-a + b\text{i})^2 = a^2 - b^2 - 2ab\text{i}$ since $b > a$, z^2 is in bottom left quadrant.



16) a)



The area is $4 \times 4 = 16$ units^2

16) b)

$-1 + 3\text{i}$ and $-1 - \text{i}$ or $7 - \text{i}$ and $7 + 3\text{i}$

17) a)

$r = \sqrt{a^2 + b^2}$

17) b)

$zz^* = (a + b\text{i})(a - b\text{i}) = a^2 + b^2 = r^2$ as req.

18) a)

$(a + b\text{i})(c + d\text{i}) = ac - bd + (bc + ad)\text{i}$

18) b)

$p = \sqrt{a^2 + b^2}$ and $q = \sqrt{c^2 + d^2}$

so $pq = \sqrt{(a^2 + b^2)(c^2 + d^2)}$

$= \sqrt{a^2c^2 + b^2d^2 + b^2c^2 + a^2d^2}$

$|zw| = \sqrt{(ac - bd)^2 + (bc + ad)^2}$

$= \sqrt{a^2c^2 - 2abcd + b^2d^2 + b^2c^2 + 2abcd + a^2d^2}$

$= \sqrt{a^2c^2 + b^2d^2 + b^2c^2 + a^2d^2}$

$\therefore |zw| = pq$ as req.

19) a)

$\text{i}z = \text{i}(2 - 3\text{i}) = 3 + 2\text{i}$

19) b)

$(1 + \text{i})z = (1 + \text{i})(2 - 3\text{i}) = 2 - 3\text{i} + 2\text{i} + 3 = 5 - \text{i}$

19) c)

Distance between each of the points:

$|z - 0| = |2 - 3\mathrm{i}| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$

$|0 - \mathrm{i}z| = |0 - 3 - 2\mathrm{i}| = |-3 - 2\mathrm{i}| = \sqrt{(-3)^2 + (-2)^2} = \sqrt{13}$

$|\mathrm{i}z - (1 + \mathrm{i})z| = |3 + 2\mathrm{i} - 5 + \mathrm{i}| = |-2 + 3\mathrm{i}| = \sqrt{(-2)^2 + 3^2} = \sqrt{13}$

$|(1 + \mathrm{i})z - z| = |5 - \mathrm{i} - 2 + 3\mathrm{i}| = |3 + 2\mathrm{i}| = \sqrt{3^2 + 2^2} = \sqrt{13}$

All the lengths are the same.

Since the transformation from z to $\mathrm{i}z$ is a rotation of $90°$ anti-clockwise about the origin, one of the angles must be $90°$ so it must be a square.

19) d)

$\sqrt{13} \times \sqrt{13} = 13$ units^2

20) a)

$\mathrm{i}z = \mathrm{i}(a + b\mathrm{i}) = -b + a\mathrm{i}$

20) b)

$(1 + \mathrm{i})z = (1 + \mathrm{i})(a + b\mathrm{i}) = a + b\mathrm{i} + a\mathrm{i} - b = (a - b) + (a + b)\mathrm{i}$

20) c)

Distance between each of the points:

$|z - 0| = |a + b\mathrm{i}| = \sqrt{a^2 + b^2}$

$|0 - \mathrm{i}z| = |0 + b - a\mathrm{i}| = |b - a\mathrm{i}| = \sqrt{b^2 + (-a)^2} = \sqrt{a^2 + b^2}$

$|-b + a\mathrm{i} \ - (a - b) - (a + b)\mathrm{i}| = |-b + a\mathrm{i} \ - a + b - a\mathrm{i} - b\mathrm{i}|$

$= |\ -a - b\mathrm{i}| = \sqrt{(-a)^2 + (-b)^2} = \sqrt{a^2 + b^2}$

$|(a - b) + (a + b)\mathrm{i} - a - b\mathrm{i}| = |a - b + a\mathrm{i} + b\mathrm{i} - a - b\mathrm{i}| = |-b + a\mathrm{i}| =$

$\sqrt{(-b)^2 + a^2} = \sqrt{a^2 + b^2}$

All the lengths are the same.

Since the transformation from z to $\mathrm{i}z$ is a rotation of $90°$ anti-clockwise about the origin, one of the angles must be $90°$ so it must be a square.

20) d)

$\sqrt{a^2 + b^2} \times \sqrt{a^2 + b^2} = (a^2 + b^2)$ units^2

21) a)

$\mathrm{i}z = \mathrm{i}(a + b\mathrm{i}) = -b + a\mathrm{i}$

So $|OP| = \sqrt{a^2 + b^2}$ and $|OQ| = \sqrt{(-b)^2 + a^2} = \sqrt{a^2 + b^2}$

21) b)

Geometrically, multiplying by i rotates a complex number 90° anti-clockwise about the origin on the Argand diagram so OP is perpendicular to OQ

Alternative Method:

gradient of line from origin to z is $\frac{b}{a}$

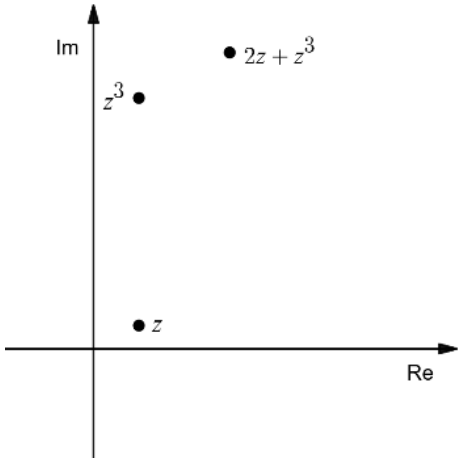
gradient of line from origin to $\mathrm{i}z = -b + a\mathrm{i}$ is $\frac{a}{-b}$

Since $\frac{b}{a} \times \frac{a}{-b} = -1$ they are perpendicular, so OP is perpendicular to OQ

22) a)

$z = 2 + \mathrm{i}$ gives $z^3 = (2 + \mathrm{i})^3 = 8 + 12\mathrm{i} - 6 - \mathrm{i} = 2 + 11\mathrm{i}$

$2z + z^3 = 2(2 + \mathrm{i}) + 2 + 11\mathrm{i} = 6 + 13\mathrm{i}$



22) b)

$|OA| = \sqrt{2^2 + 1^2} = \sqrt{5}$

22) c)

$|BC| = |6 + 13\mathrm{i} - 2 - 11\mathrm{i}| = |4 + 2\mathrm{i}| = \sqrt{4^2 + 2^2} = \sqrt{20} = 2\sqrt{5}$

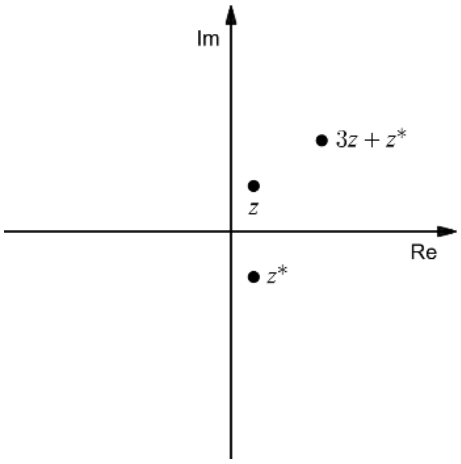
22) d)

Line segment $|BC|$ is twice the length of $|OA|$

23) a)

$z = 1 + 2\text{i}$ gives $z^* = 1 - 2\text{i}$

$3z + z^* = 3(1 + 2\text{i}) + 1 - 2\text{i} = 4 + 4\text{i}$



23) b)

$|AC| = \sqrt{3^2 + 4^2} = \sqrt{13}$

23) c)

$|BC| = |4 + 4\text{i} \ - 1 + 2\text{i}| = |3 + 6\text{i}| = \sqrt{3^2 + 6^2} = \sqrt{45} = 3\sqrt{5}$

23) d)

Area of $ABC = \frac{1}{2} \times 4 \times 3 = 6$ units^2

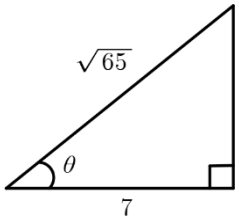
Alternative Method:

$|AB| = 4$

$4^2 = \left(3\sqrt{5}\right)^2 + \left(\sqrt{13}\right)^2 - 2 \times \left(3\sqrt{5}\right)\left(\sqrt{13}\right)\cos\theta$

$16 = 45 + 13 - 6\sqrt{65}\cos\theta$

$\cos\theta = \frac{7}{\sqrt{65}}$



So $\sin\theta = \frac{4}{\sqrt{65}}$

Area of $ABC = \frac{1}{2} \times 3\sqrt{5} \times \sqrt{13} \times \frac{4}{\sqrt{65}} = 6$ units^2

Alternative Method:

$|AB| = 4$

$s = \frac{4+3\sqrt{5}+\sqrt{13}}{2}$

Area $= \sqrt{\frac{4+3\sqrt{5}+\sqrt{13}}{2}\left(\frac{4+3\sqrt{5}+\sqrt{13}}{2} - 3\sqrt{5}\right)\left(\frac{4+3\sqrt{5}+\sqrt{13}}{2} - \sqrt{13}\right)\left(\frac{4+3\sqrt{5}+\sqrt{13}}{2} - 4\right)} = 6$ units^2

24) a)

$z = 1 + \mathrm{i}$

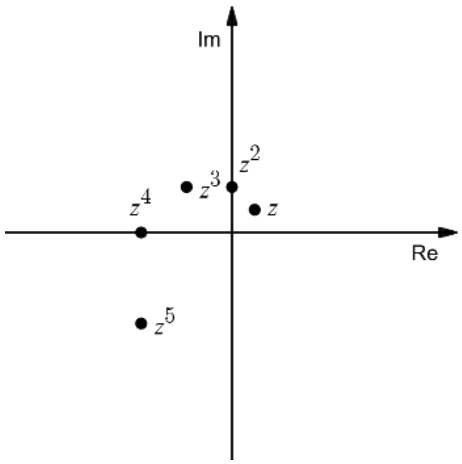
$z^2 = (1+\mathrm{i})(1+\mathrm{i}) = 1 + \mathrm{i} + \mathrm{i} - 1 = 2\mathrm{i}$

$z^3 = 2\mathrm{i}(1+\mathrm{i}) = -2 + 2\mathrm{i}$

$z^4 = (2\mathrm{i}-2)(1+\mathrm{i}) = 2\mathrm{i} - 2 - 2 - 2\mathrm{i} = -4$

$z^5 = -4(1+\mathrm{i}) = -4 - 4\mathrm{i}$

24) b)



24) c)

$|z| = |1+\mathrm{i}| = \sqrt{1^2+1^2} = \sqrt{2}$

$|z^2| = |2\mathrm{i}| = 2$

$|z^3| = |-2+2\mathrm{i}| = \sqrt{(-2)^2+2^2} = \sqrt{8} = 2\sqrt{2}$

$|z^4| = |-4| = 4$

$|z^5| = |-4-4\mathrm{i}| = \sqrt{(-4)^2+(-4)^2} = \sqrt{32} = 4\sqrt{2}$

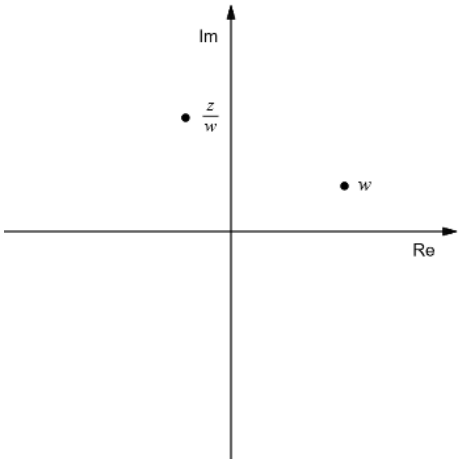
24) d)

$|z^n| = \left(\sqrt{2}\right)^n = 2^{\frac{n}{2}}$

25) a)

$\frac{z}{w} = \frac{-20+21\mathrm{i}}{5+2\mathrm{i}} \times \frac{5-2\mathrm{i}}{5-2\mathrm{i}} = \frac{-58+145\mathrm{i}}{25+4} = -2+5\mathrm{i}$ as req.

25) b)



25) c)

$w = 5+2\mathrm{i}$ and $\frac{z}{w} = -2+5\mathrm{i}$

Since $w\mathrm{i} = \mathrm{i}(5+2\mathrm{i}) = -2+5\mathrm{i} = z$

Geometrically, multiplying by i rotates a complex number $90°$ anti-clockwise about the origin on the Argand diagram so they are perpendicular, the points w and $\frac{z}{w}$ form a right-angled triangle.

Alternative Method:

gradient of line from origin to w is $\frac{5}{2}$

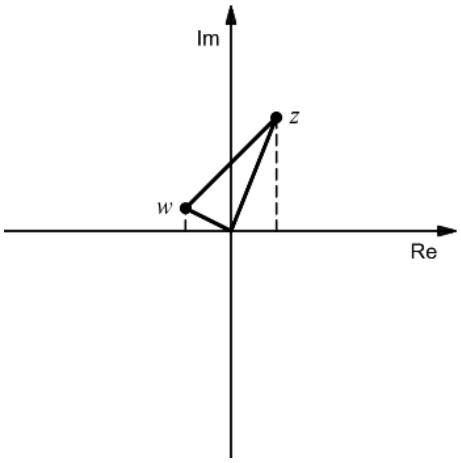
gradient of line from origin to $\frac{z}{w}$ is $\frac{-2}{5}$

Since $\frac{5}{2} \times \frac{-2}{5} = -1$ they are perpendicular, the points w and $\frac{z}{w}$ form a right-angled triangle.

25) d)

Area $= \frac{1}{2}\sqrt{(-2)^2+5^2}\sqrt{5^2+2^2} = \frac{29}{2}$ units^2

26)



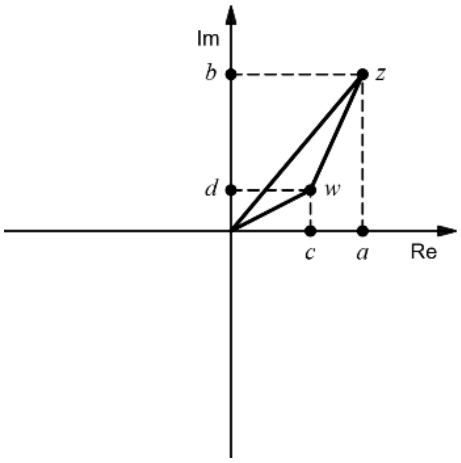
Area of trapezium: $\frac{1}{2}(1+5)\times 4 = 12$

Area of triangle 1: $\frac{1}{2}\times 2\times 1 = 1$

Area of triangle 2: $\frac{1}{2}\times 2\times 5 = 5$

$\therefore$ Area is $12-1-5 = 6$ units^2 as req.

27)



Large triangle: $\frac{1}{2}ab$ and small triangle: $\frac{1}{2}cd$

Trapezium: $\frac{1}{2}(b+d)(a-c)$

So area wanted:

$\frac{1}{2}ab-\frac{1}{2}cd-\frac{1}{2}(b+d)(a-c)$

$=\frac{1}{2}ab-\frac{1}{2}cd-\frac{1}{2}(ba-bc+da-dc)$

$=\frac{1}{2}ab-\frac{1}{2}cd-\frac{1}{2}ab+\frac{1}{2}cb-\frac{1}{2}ad+\frac{1}{2}cd$

$=\frac{1}{2}bc-\frac{1}{2}ad$

$=\frac{1}{2}(bc-ad)$

To keep the area positive: $\frac{1}{2}|bc-ad|$

But since $zw^*=(a+b\mathrm{i})(c-d\mathrm{i})=ac+bd+(bc-ad)\mathrm{i}$

$\mathrm{Im}(zw^*)=bc-ad$

So that area $=\frac{1}{2}|\mathrm{Im}(zw^*)|$ as req.

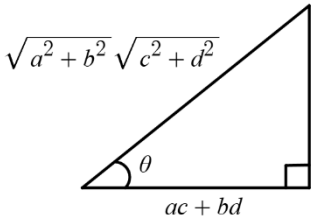
Alternative Method:

Let $z=a+b\mathrm{i}$ and $w=c+d\mathrm{i}$

$zw^*=(a+b\mathrm{i})(c-d\mathrm{i})=ac+bd+(bc-ad)\mathrm{i}$ so $\mathrm{Im}(zw^*)=bc-ad$

$|z^*|=\sqrt{a^2+b^2}$ and $|w|=\sqrt{c^2+d^2}$

$\cos\theta=\frac{ac+bd}{\sqrt{a^2+b^2}\sqrt{c^2+d^2}}$



So that the opposite side is given by:

$\sqrt{(a^2+b^2)(c^2+d^2)-(ac+bd)^2}$

$=\sqrt{a^2c^2+a^2d^2+b^2c^2+b^2d^2-a^2c^2-2abcd-b^2d^2}$

$=\sqrt{a^2d^2+b^2c^2-2abcd}$

$=\sqrt{(ad-bc)^2}$

$=ad-bc$

$\sin\theta=\frac{ad-bc}{\sqrt{a^2+b^2}\sqrt{c^2+d^2}}$

Area: $\frac{1}{2}\sqrt{a^2+b^2}\sqrt{c^2+d^2}\times\frac{ad-bc}{\sqrt{a^2+b^2}\sqrt{c^2+d^2}}$

$=\frac{1}{2}(ad-bc)$

$=\frac{1}{2}|ad-bc|$

$=\frac{1}{2}|bc-ad|$

$=\frac{1}{2}|\mathrm{Im}(zw^*)|$ as req.